



Banking Analytics Dashboard Report

1. Project Overview

The Banking Analytics Dashboard is an end-to-end data analytics project developed to analyze customer banking behavior, lending activities, deposit portfolios, and financial product utilization. The project combines Python-based Exploratory Data Analysis (EDA) with Power BI dashboarding to transform raw banking data into actionable business insights.

2. Business Objective

The banking industry generates large volumes of transactional and customer data. Without proper analysis, it becomes difficult to identify customer trends, monitor financial performance, and evaluate lending and deposit portfolios.

This project addresses these challenges by:

- Analyzing customer financial behavior.
- Monitoring lending and deposit performance.
- Developing business-focused KPIs.
- Creating an interactive reporting solution.
- Supporting strategic decision-making.

3. Dataset Overview

Metric	Details
Total Customers	3,000
Lending Products	Bank Loans, Business Lending, Credit Card Balance
Deposit Products	Bank Deposits, Savings, Checking, Foreign Currency
Customer Attributes	Age, Gender, Nationality, Occupation
Relationship Types	Retail, Institutional, Private Bank, Commercial

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
# Generate descriptive statistics for the dataframe
df.describe()
```

	Age	Location ID	Estimated Income	Superannuation Savings	Amount of Credit Cards	Credit Card Balance	Bank Loans	Bank Deposits	Checking Accounts	Saving Accounts
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3.000000e+03	3.000000e+03	3.000000e+03	3.000000e+03
mean	51.039667	21563.323000	171305.034263	25531.599673	1.463667	3176.206943	5.913862e+05	6.715602e+05	3.210929e+05	2.329084e+05
std	19.854760	12462.273017	111935.808209	16259.950770	0.676387	2497.094709	4.575570e+05	6.457169e+05	2.820796e+05	2.300078e+05
min	17.000000	12.000000	15919.480000	1482.030000	1.000000	1.170000	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
25%	34.000000	10803.500000	82906.595000	12513.775000	1.000000	1236.630000	2.396281e+05	2.044004e+05	1.199475e+05	7.479440e+04
50%	51.000000	21129.500000	142313.480000	22357.355000	1.000000	2560.805000	4.797934e+05	4.633165e+05	2.428157e+05	1.640866e+05
75%	69.000000	32054.500000	242290.305000	35464.740000	2.000000	4522.632500	8.258130e+05	9.427546e+05	4.348749e+05	3.155750e+05
max	85.000000	43369.000000	522330.260000	75963.900000	3.000000	13991.990000	2.667557e+06	3.890598e+06	1.969923e+06	1.724118e+06

Foreign Currency Account	Business Lending	Properties Owned	Risk Weighting	BRId	GenderId	IAId
3000.000000	3.000000e+03	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000
29883.529993	8.667598e+05	1.518667	2.249333	2.559333	1.504000	10.425333
23109.924010	6.412303e+05	1.102145	1.131191	1.007713	0.500067	5.988242
45.000000	0.000000e+00	0.000000	1.000000	1.000000	1.000000	1.000000
11916.542500	3.748251e+05	1.000000	1.000000	2.000000	1.000000	5.000000
24341.190000	7.113147e+05	2.000000	2.000000	3.000000	2.000000	10.000000
41966.392500	1.185110e+06	2.000000	3.000000	3.000000	2.000000	15.000000
124704.870000	3.825962e+06	3.000000	5.000000	4.000000	2.000000	22.000000

- **Banking Relationship Analysis:** Understand customer segmentation across banking relationship categories.

Key Insight: The customer base is distributed across multiple banking relationship types providing opportunities for targeted product offerings and personalized banking services.

- **Deposit Portfolio Analysis:** Analyze customer deposit behavior.

Key Insight: Deposits are not uniformly distributed among customers, indicating the presence of high-value and low-value deposit segments.

- **Loan Portfolio Analysis:** Analyze customer lending exposure.

Key Insight: Loan balances vary significantly across customers, highlighting different borrowing patterns and customer financial profiles.

- **Correlation Analysis:** Identify relationships among financial variables.

Key Insight:

- Savings Accounts positively contribute to total deposits.
- Checking Accounts show strong association with deposit balances.
- Foreign Currency Accounts contribute to overall customer value.
- Lending products demonstrate relationships with income and deposit behavior.

- **Cleaning & Preparation:**

- Standardized column names to snake_case.
- Validated financial balances to ensure no negative values where inappropriate.
- Feature Engineering: Categorized customers into Income Bands (Low, Med, High).

5. KPI Development

To evaluate overall banking performance and customer financial behavior, several Key Performance Indicators (KPIs) were developed using DAX in Power BI. These KPIs provide a high-level view of lending activities, deposit portfolios, customer value, and banking efficiency.

- **Loan-to-Deposit Ratio (LDR): 116%**
 - *Insight:* An LDR above 100% indicates that the bank is lending more than it is taking in as deposits, which may suggest a need for higher liquidity management.

KPI Summary

KPI	Description	Business Significance
Total Customers	Total number of unique customers in the dataset	Helps measure the size of the customer base and overall business reach
Total Loans	Sum of Bank Loans, Business Lending, and Credit Card Balances	Represents the bank's total lending exposure
Total Deposits	Sum of all deposit-related products including savings, checking, foreign currency accounts, and bank deposits	Measures the total funds held by customers within the bank
Bank Loans	Total value of traditional bank loans	Used to analyze the bank's core lending portfolio
Business Lending	Total value of loans provided to businesses	Helps evaluate commercial lending performance
Bank Deposits	Total amount deposited in banking accounts	Indicates the bank's deposit strength and liquidity position
Savings Balance	Total balance held in savings accounts	Measures customer saving behavior and long-term deposits
Checking Balance	Total balance held in checking accounts	Reflects customers' day-to-day banking activity
Credit Card Balance	Total outstanding credit card balances	Helps monitor revolving credit exposure
Foreign Currency Balance	Total value held in foreign currency accounts	Indicates customer participation in foreign currency products
Loan-to-Deposit Ratio (LDR)	Total Loans ÷ Total Deposits	Measures lending activity relative to available deposits and serves as an important liquidity indicator
Average Deposit per Customer	Total Deposits ÷ Total Customers	Measures the average deposit value contributed by each customer

HOME

LOAN ANALYSIS

DEPOSIT ANALYSIS

SUMMARY

2013

2018

2019

2020

2021

Banking Relationship

Commercial

Institutional

Private Bank

Retail

Gender

Female

Male

Investment Advisor

All

Total Customers

3000

Total Loans

4.38bn

Bank Loan

1.77bn

Business Lending

2.60bn

Total Deposits

3.77bn

Bank Deposit

2.01bn

Checking Balance

963.28M

Savings Balance

698.73M

Foreign Currency Balance

89.65M

Loan to Deposit Ratio

116%

Avg Loan / Customer

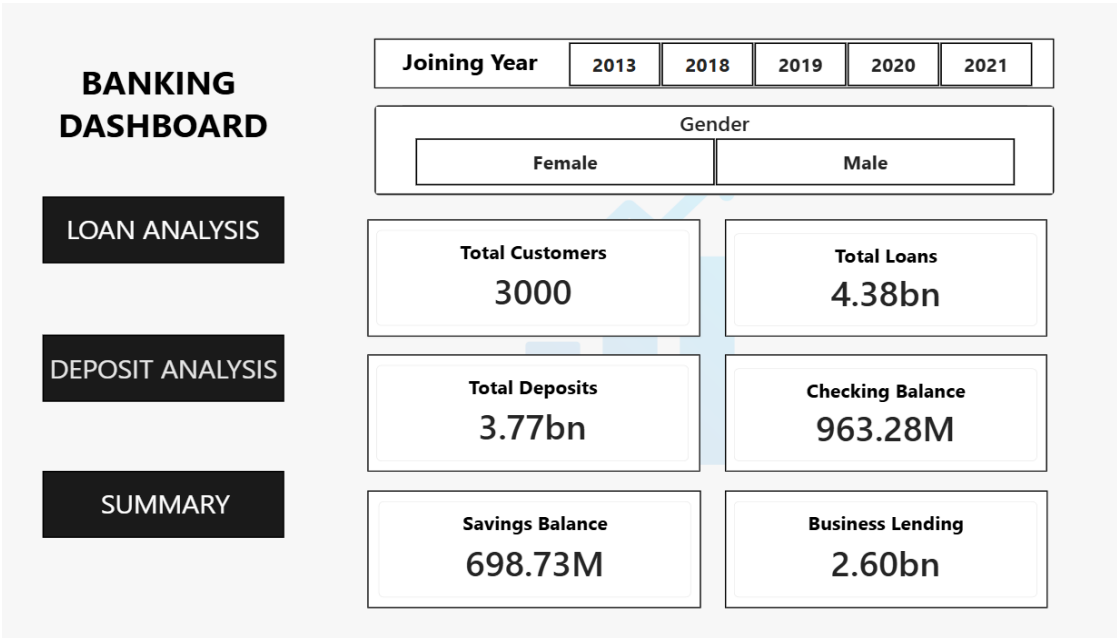
1.46M

Avg Deposit / Customer

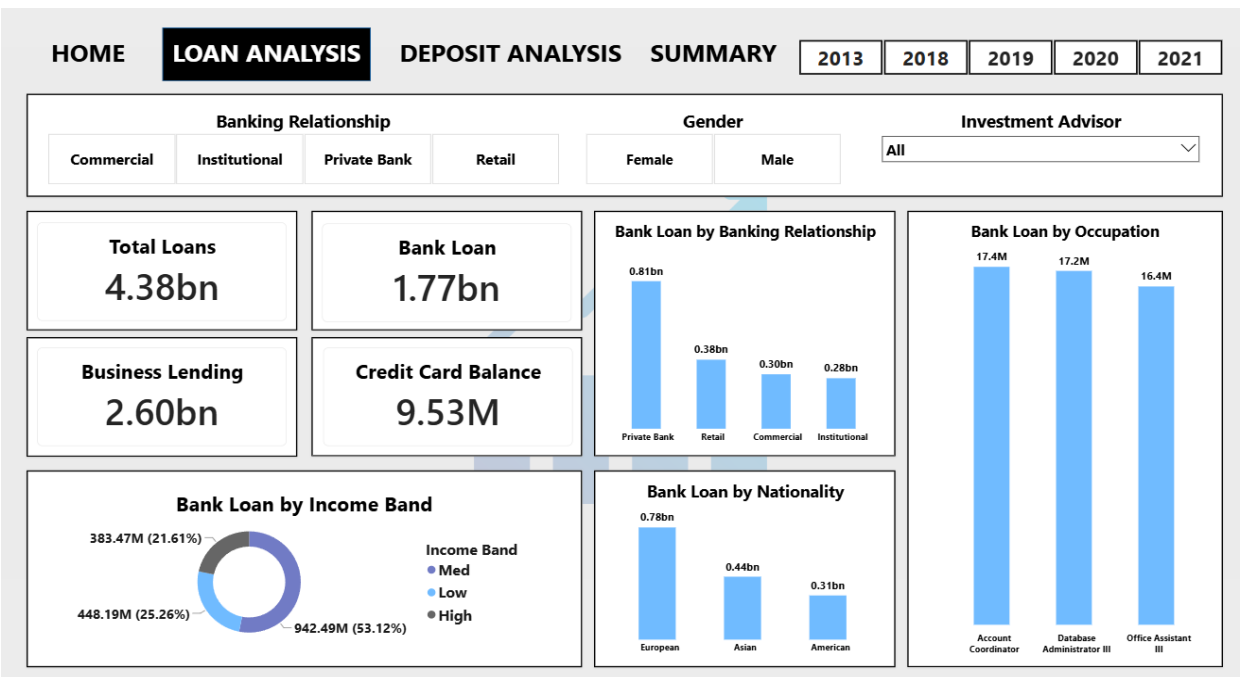
1.28M

6. Dashboard Development & Visualization

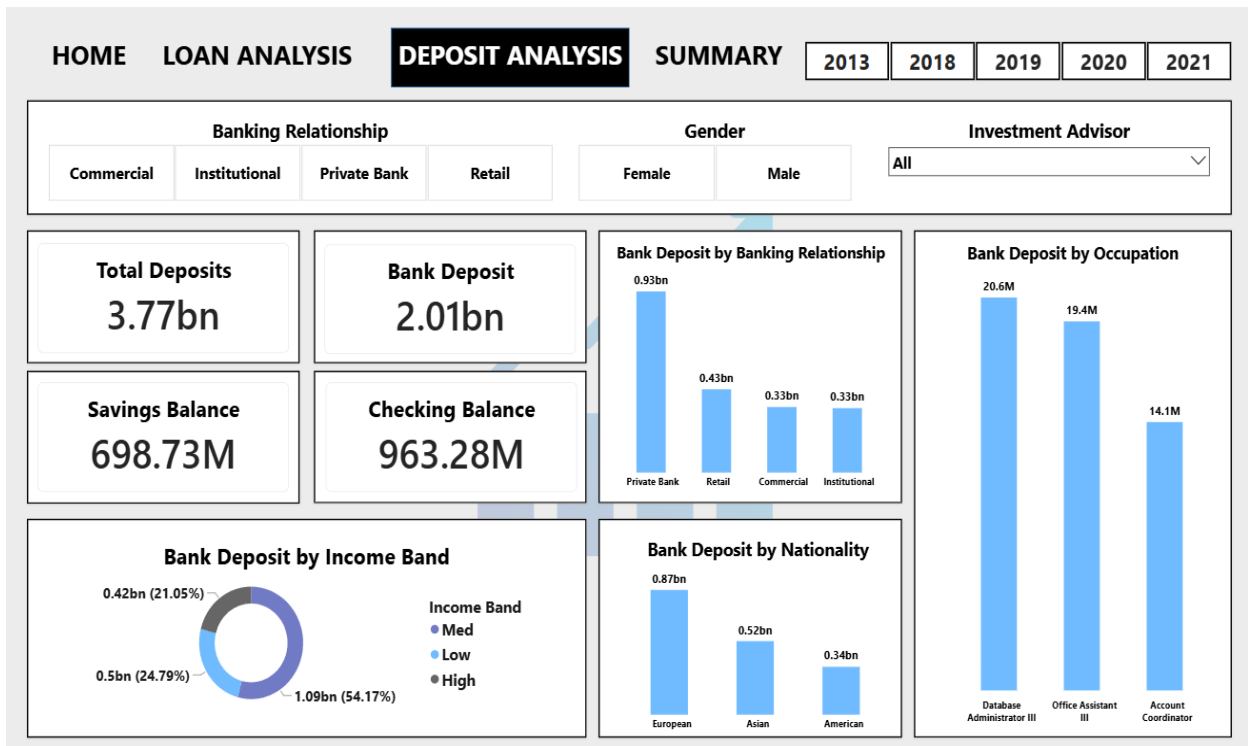
1. **Home Page:** High-level summary of total customers and core financial balances.



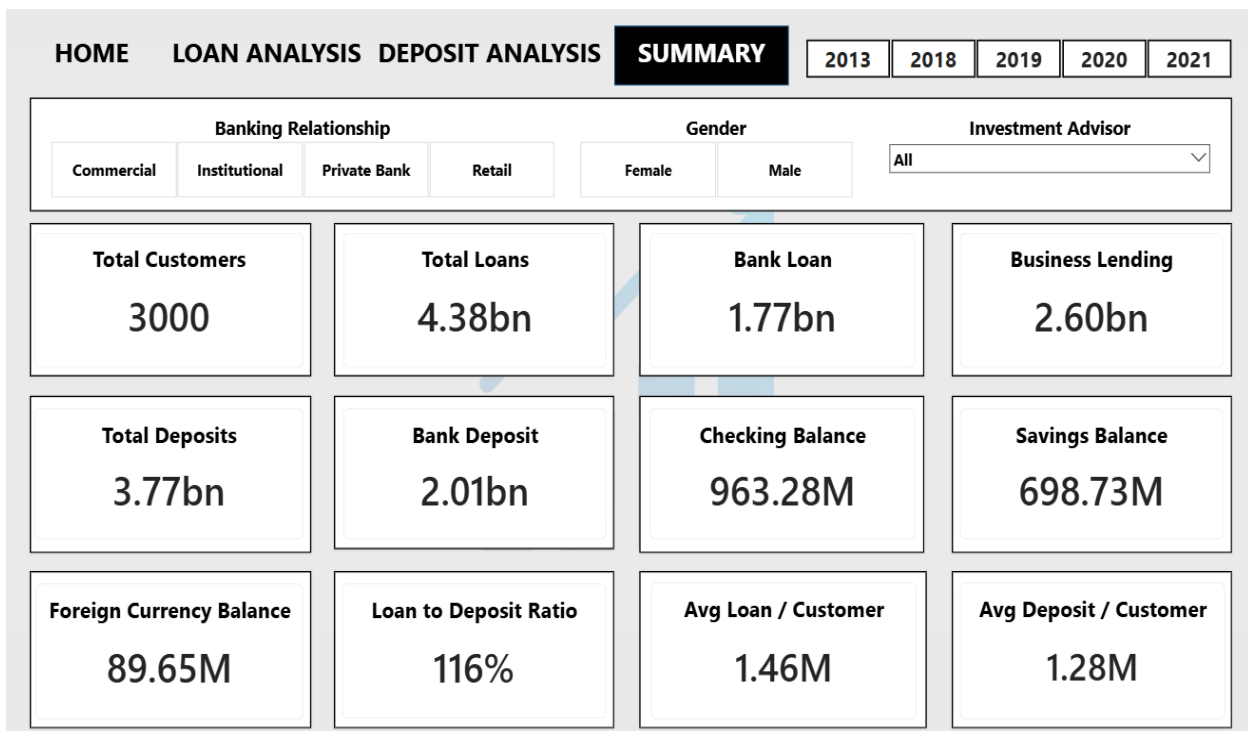
2. **Loan Analysis:** Detailed breakdown of lending by demographics and relationship types.



3. **Deposit Analysis:** Visualizations of saving vs. checking trends and income-based deposit clusters.



4. **Summary Dashboard:** Advanced metrics including LDR and average balances per customer.



7. Business Recommendations

- 1. Focus on Growing Customer Deposits:** The bank's loan portfolio is larger than its deposit base, indicating an opportunity to strengthen deposit growth. Introducing attractive savings products and customer loyalty programs can help improve liquidity and support future lending activities.
- 2. Strengthen Relationships with High-Value Customers:** Private Banking customers contribute significantly to both deposits and loans. Providing personalized financial services and wealth management solutions can improve customer retention and increase long-term profitability.
- 3. Increase Adoption of Credit Card Products:** Credit card balances represent a relatively small portion of the overall lending portfolio. Targeted promotions and bundled offers can encourage existing customers to adopt additional credit products.
- 4. Develop Occupation-Based Banking Solutions:** Certain occupations show higher engagement with banking products and services. Designing customized loan and deposit offerings for these customer groups can enhance customer satisfaction and business growth.
- 5. Leverage Customer Insights for Personalized Marketing:** Customer demographics and banking relationship data can be used to create targeted marketing campaigns. Personalized product recommendations can improve engagement, increase product adoption, and strengthen customer relationships.

8. Conclusion

This project showcases the complete journey of a data analytics solution—from data preparation and exploratory analysis to KPI creation and dashboard development. The insights generated from the analysis help better understand customer behavior, lending performance, and deposit management. Overall, the project highlights how data can be used to support smarter business decisions and improve banking operations.

9. Tools & Technologies Used

Category	Tools
Programming	Python
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Business Intelligence	Power BI
Data Modeling	DAX
Domain	Banking Analytics

